

Real-Time Meteorological Data Sources for AQI ML Project

Introduction

This document summarizes the top real-time meteorological data sources suitable for training and operating machine learning (ML) models for Air Quality Index (AQI) prediction, with a focus on India and global coverage. It covers official government APIs, open-data platforms, and commercial providers, along with practical guidance on integrating these sources into an ML pipeline.

Key Meteorological Features for AQI Modeling

ML models for AQI typically use the following meteorological features as inputs:

- Temperature (near-surface, multi-level)
- Relative humidity / dew point
- Wind speed and wind direction
- Atmospheric pressure (e.g., sea-level pressure)
- Precipitation (rain/snow amount)
- Solar radiation / sunlight
- Boundary layer height / vertical mixing
- Cloud cover and visibility
- Turbulence / stability indicators

Top Real-Time Meteorological Data Sources

1. India Meteorological Department (IMD) – Official India-Specific Source

IMD provides authoritative, India-specific real-time and forecast weather data via its API platform. It is the primary recommended source for projects focused on Indian cities such as Pune.

Aspect	Details
Base URL	https://api.imd.gov.in/
Key Endpoints	Current weather: <code>/api/v1/current_wx</code> AWS data: <code>/api/v1/aws_data</code> District nowcast: <code>/api/v1/districtnowcast</code>
Variables	Temperature, humidity, wind speed/direction, pressure, rainfall, etc.
Update Frequency	Typically hourly; varies by product
Access	Registration and API key required; check terms and rate limits
Best For	Official, India-specific meteorological data for AQI modeling

2. Open-Meteo – Global, Free, No API Key (Non-Commercial)

Open-Meteo is a free, open-source weather API with global coverage, high resolution, and no API key requirement for non-commercial use. It is excellent for prototyping and production if the license fits your use

case.

Aspect	Details
Base URL	https://open-meteo.com/
Key Endpoint	https://api.open-meteo.com/v1/forecast
Variables	Temperature, relative humidity, wind speed/direction, surface pressure, precipitation, cloud cover, solar
Update Frequency	Every 1–6 hours depending on region; hourly interpolated output
Access	No API key required; ~10k calls/day free for non-commercial
Best For	Easy integration, global coverage, rich feature set for ML

3. Other Notable Meteorological Data Sources

Depending on your requirements (commercial use, SLAs, additional features), consider these alternatives:

- Xweather: Global weather API with real-time and historical data; ML-ready features; commercial tiers available.
- ECMWF via Open-Meteo: High-resolution global forecasts (e.g., IFS/AIFS) accessible through Open-Meteo; suitable for advanced modeling.
- National/regional services: E.g., NOAA (USA), DMI (Denmark), Met Office (UK) – often accessible via their own APIs or through aggregators.
- Commercial providers: E.g., IQAir AirVisual (includes weather + AQI), Plume Labs, and others offering paid APIs with support and SLAs.

Complementary Real-Time AQI / Pollutant Data Sources

For labels and joint modeling, combine meteorological data with real-time pollutant measurements from:

- CPCB (India): Central Pollution Control Board provides real-time AQI and pollutant data via dashboard and API (data.gov.in).
- OpenAQ: Global aggregator of official air quality data; API and bulk downloads; includes Indian stations.
- AQICN: Real-time AQI and station-level data for many countries, including India; API available.
- IQAir AirVisual: Commercial API with real-time AQI, PM2.5/PM10, and weather data; useful for co-located features.

Integration into ML Pipeline

A typical real-time ML pipeline for AQI prediction includes:

1. Select primary meteorological source (e.g., IMD for India, Open-Meteo for global).
2. Define feature set (temperature, humidity, wind, pressure, etc.) and polling frequency (e.g., every 15–60 minutes).
3. Implement a service to fetch and store latest meteorological and pollutant snapshots (database/cache).
4. Feed these features into your trained model (e.g., Random Forest, XGBoost, neural network) for inference.
5. Periodically retrain using historical data (IMD archives, CPCB historical, OpenAQ bulk data) to adapt to seasonal changes.

Example: Open-Meteo API Call for Pune

The following HTTP request retrieves hourly meteorological features for Pune (latitude 18.52, longitude 73.85):

```
GET https://api.open-meteo.com/v1/forecast?latitude=18.52&longitude=73.85&hourly=temperature_2m,relativehumidity_2m,windspeed_10m,winddirection_10m,surface_pressure,precipitation,cloudcover&forecast_days=1
```

Use the latest hour from the 'hourly' arrays as your current meteorological input to the model.

Recommended Stack for Pune-Based AQI Project

Component	Recommended Source(s)
Meteorology (primary)	IMD API (current_wx, aws_data) for Pune stations
Meteorology (fallback/extra)	Open-Meteo (global, no key, rich variables)
Pollutants / AQI (primary)	CPCB AQI API via data.gov.in for Pune CAAQMS stations
Pollutants / AQI (secondary)	AQICN or OpenAQ for easier access and broader coverage

Note

Always verify API terms, rate limits, and licensing (especially for commercial use). For IMD and CPCB APIs, registration and API keys are typically required. Open-Meteo is free for non-commercial use without an API key.